

BACKGROUND	
Title/Citation	Cannon C, Blazing M, Giugliano R, et al. Ezetimibe added to Statin therapy after Acute Coronary Syndromes. NEJM.org June 18, 2015 Accessed February 16, 2024 https://www.nejm.org/doi/full/10.1056/NEJMoa1410489
Background	- Hospitalized patients for acute coronary syndrome and treating with Statin therapy reduces low-density lipoprotein (LDL) cholesterol levels and the risk of cardiovascular events. - Ezetimibe, a non-statin drug that reduces intestinal cholesterol absorption to lower LDL level than previous target will reduce the rate of cardiovascular events and rehospitalization. - Recommendation by AHA as Class Ia with high intensity statin therapy with Ezetimibe before starting PCSK9 inhibitor.
GENERAL STUDY OVERVIEW	
Purpose	Addition of Ezetimibe a non-statin drug added to high intensity statin therapy for LDL lowering treatment in patients with Acute coronary syndrome will receive more benefit and reduce the risk of rehospitalization or MI.
Funding	Merck (IMPROVE-IT Clinical Trial number NCT00202878)
Design	Double-blind, randomized 1:1 ratio trial with median follow-up of 6 years
METHODS	
Inclusion criteria	- Men and women at least 50 years of age. - Hospitalized for acute coronary syndrome within the preceding 10 days. - Minimum LDL cholesterol levels of 50 mg per deciliter. - With receiving lipid-lowering therapy maximum 100 mg per deciliter. - Not receiving lipid-lowering therapy max 125 mg per deciliter.
Exclusion criteria	- Planned coronary-artery bypass grafting for the acute coronary syndrome event. - Creatinine clearance of less than 30 ml per minute. - Active liver disease, or use of statin therapy that had LDL cholesterol-lowering potency greater than 40 mg of simvastatin.
Intervention	- Randomly assigned patient with 1:1 ratio of Once daily, either simvastatin (at a dose of 40 mg) plus Ezetimibe (at a dose of 10 mg) (simvastatin–ezetimibe group) or simvastatin (at a dose of 40 mg) plus placebo (simvastatin-monotherapy group). - The number of total patients was 18,144 with median follow up was 6 years.
Primary outcome	Death from cardiovascular cause, major coronary event or nonfatal stroke.
Secondary outcomes	- Death from any cause, major coronary event or nonfatal stroke. - Death from coronary heart disease, nonfatal MI, urgent coronary revascularization ≥ 30 days. - Death from cardiovascular causes, nonfatal MI, hospitalization for unstable angina, all revascularization ≥ 30 days, nonfatal stroke.
Procedures	- Patients received standard medical and interventional treatment for acute coronary syndrome and were randomly assigned in a 1:1 ratio and in a double-blind fashion to receive, either Ezetimibe 10 mg with addition of Simvastatin at a dose of 40 mg or standard LDL lowering care of Simvastatin 40 mg dose plus placebo. - Patients had follow-up visits at 30 days, at 4 months, and every 4 months thereafter. - Blood samples were obtained at randomization, at 1, 4, 8, and 12 months, and yearly thereafter for 6 years.
Statistical analyses	5250 events were required for 90% power to detect 9.375% relative lowering risk with Simvastatin and Ezetimibe combination therapy with safety analyses for intention to treat. - A P-value of 0.05 or less without adjustment was used and Estimated Hazard ratios were associated with 95% Confidence intervals was used to compare Simvastatin – Ezetimibe and Simvastatin monotherapy with Coz proportional hazards model. - Kaplan-Meier failure rates method was used to rate the 7-year event rate.
RESULTS	

Baseline characteristics	18,144 patients were randomized enrolled at 1147 sites in 39 countries. 9077 were assigned to Simvastatin monotherapy group. 9067 was assigned to Simvastatin – Ezetimibe group. - Average age of patients was 64 years and 24% were women and 27% had diabetes mellitus. 88% had coronary angiography and 70% had percutaneous coronary intervention. - Simvastatin 80mg was added to 27% of Monotherapy group and 6% of Combination therapy during the study period. - Median of 6 years 42% patients discontinued study in both groups.
Summary of outcomes	Primary: Patients were at Simvastatin – Ezetimibe combo therapy Kaplan-Meier event rate was 32.7% compared to Simvastatin monotherapy (34.7%) with the absolute risk reduction of 2%. (hazard ratio 0.936; 95% CI, 0.89 to 0.99; P=0.016) Secondary: There was no significant high risk of hemorrhagic stroke with Simvastatin – Ezetimibe group than Simvastatin monotherapy. - The rate of death from Cardiovascular causes were similar in both treatment groups.
AUTHORS' DISCUSSION & CONCLUSIONS	
Author's Discussion	- Ezetimibe, a non-statin agent is added to the standard of therapy led to more successful LDL cholesterol lowering by 24% - Leads to a significantly lower risk of cardiovascular events than the Statin monotherapy.
Author's conclusions	Addition of Ezetimibe to high statin therapy observed with significant reduction of myocardial infarction and ischemic stroke.
PRESENTER'S DISCUSSION & CONCLUSIONS	
Study strengths	- Appropriately powered, designed to minimize confounding (with blinding, randomization, and inclusion/exclusion criteria). - Baseline medication use, age, and level of LDL 50-100mg/dL is similar between groups. - Ezetimibe combo therapy resulted in improved LDL lowering vs. Simvastatin monotherapy (53.7mg/dL VS. 69.5mg/dL)
Study limitations	- The patients with acute coronary syndrome were mostly white males as part of the study and the time length of study from 6 years to 7 years was adjusted during the study. - Another limitation we can see was use of 40 and 80mg of Simvastatin was used according to the current guidelines therapy according to LDL levels of patients at the time of enrolment.
Conclusion and Application to patient care	When added to statin therapy, ezetimibe resulted in incremental lowering of LDL cholesterol levels and improved cardiovascular outcomes.

